

RAVIKANTH DULAM

Senior DevOps & MLOps Engineer

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PROFESSIONAL SUMMARY

Senior DevOps & MLOps Engineer with 6+ years of hands-on experience architecting, automating, and scaling cloud-native infrastructure and end-to-end MLOps pipeline development across AWS, Azure, GCP, and Alibaba Cloud (Ali). Recognized for slashing deployment times by 50%, sustaining 99.9% production uptime, and engineering production-grade MLOps platforms that serve real-time ML inference at scale. Deep expertise in Kubernetes (AKS/EKS/GKE), Terraform, Ansible, Azure DevOps, GitHub Actions, Jenkins (incl. shared libraries), Docker, Argo CD, Helm, MLflow, Py Spark, NGINX ingress/TLS termination, Hashi Corp Vault-based secrets management, and DevSecOps toolchains. Skilled in model deployment, model lifecycle management, automated model training pipelines, data drift detection, model monitoring, experiment tracking, and REST API-based ML inference deployment. Experienced owning production reliability end-to-end — RCA, on-call/support rotation, and an SLO/SLI-driven mindset — while bridging data science, software engineering, and infrastructure teams within Agile methodologies (Scrum/Kanban) to deliver secure, observable, and cost-optimized solutions at enterprise scale. Proven track record delivering high-impact solutions for global clients including Publicis Sapient, Novelis, Micron, Multiscale Technologies, and Cloud Mellow Technologies.

TECHNICAL SKILLS

CI/CD & DevOps	Azure DevOps, Azure Pipelines, Jenkins, Jenkins Shared Libraries, GitHub Actions, Argo CD, GitOps, ML Pipeline Automation, CI/CD for Machine Learning, Trunk-Based Development
Cloud – AWS	EC2, S3, VPC, IAM, ALB, ASG, ECS, Fargate, ECR, Lambda, RDS, CloudWatch, Route 53, AWS CDK (Python), AWS Secrets Manager
Cloud – Azure	AKS, Azure VMs, Azure Storage, Azure Key Vault, Azure Monitor, ARM Templates, Azure Container Registry (ACR), Azure Service Bus, Azure Log Analytics
Cloud – GCP & Alibaba Cloud	Google Kubernetes Engine (GKE), Compute Engine, Cloud Storage, Cloud IAM, Cloud Monitoring, Cloud Build; Alibaba Cloud (Ali) ECS & networking fundamentals — multi-cloud platform engineering across AWS, Azure, GCP, and Ali
MLOps & ML Infra	MLOps Pipeline Development, Azure Machine Learning, MLflow, Kubeflow Pipelines, End-to-End ML Pipelines, Model Deployment, Model Lifecycle Management, Model Versioning, Model Registry, Automated Model Promotion, Model Monitoring, Data Drift Detection, Model Performance Monitoring, Feature Stores, Experiment Tracking
Data & Processing	Py Spark, Distributed Data Processing, Feature Engineering Pipelines, Apache Kafka, Pandas, NumPy, ETL Pipelines, Data Validation
Containers & Orchestration	Docker, Kubernetes (Deployments, Services, Ingress, RBAC, HPA, VPA), Helm, AWS ECS/Fargate, AKS, GKE, Istio Service Mesh, Kustomize
Infrastructure as Code	Terraform (Modules, State Management, Workspaces), Azure ARM Templates, AWS CDK (Python), Ansible, Terragrunt
Monitoring & Observability	Prometheus, Grafana, Azure Monitor, AWS CloudWatch, ELK Stack (Elasticsearch, Logstash, Kibana), Jaeger, Loki, Alertmanager, PagerDuty, SLO/SLI Definition & Tracking
Secrets & Security / DevSecOps	Hashi Corp Vault (dynamic secrets, rotation, least-privilege access), Azure Key Vault, AWS Secrets Manager, SonarQube, OWASP ZAP, Trivy, Snyk, Checkov, IAM Policies, RBAC, Network Policies, Zero-Trust Architecture
Networking, Ingress & Edge	NGINX (reverse proxy, ingress controllers, routing, TLS/SSL termination, secure header configuration), Certificate Management, Load Balancing
API & serving	Flask, Fast API, Unicorn, Uvicorn, REST APIs, gRPC, Swagger/Open API
Reliability & Incident Management	Root Cause Analysis (RCA), Incident Response & Postmortems, On-Call/Support Rotation, SLO/SLI Mindset, Production Troubleshooting Across Linux/Unix, Kubernetes, CI/CD & Cloud Layers
Scripting & Languages	Python (Advanced, packaging & testing), Bash, Shell Scripting, YAML, JSON, HCL (Terraform)

Version Control	Git, GitHub, Bitbucket, GitLab, Branch Strategies (Git Flow, Trunk-Based), Code Review, PR Automation
Operating Systems	Linux (Ubuntu), Unix, RHEL, Windows Server, Shell Scripting
Agile & Methodologies	Agile Methodologies, Scrum, Kanban, Sprint Planning, Cross-Functional Collaboration, Backlog Grooming

PROFESSIONAL EXPERIENCE

Senior DevOps & MLOps Engineer | Multiscale Technologies Inc Dec 2019 – Dec 2025

Leading DevOps & MLOps engineering across AWS, Azure, and GCP for enterprise clients — driving infrastructure automation, MLOps pipeline development, model deployment, and security-first CI/CD within Agile delivery teams.

- Architected and owned end-to-end DevOps & MLOps pipeline development strategy across 10+ enterprise applications on AWS, Azure, and GCP, governing infrastructure provisioning, environment management, CI/CD automation, and ML pipeline orchestration for teams of 30+ engineers working in Agile (Scrum) sprints.
- Engineered and scaled multi-stage CI/CD pipelines using Azure Pipelines, GitHub Actions, and Jenkins (with reusable shared libraries) across 15+ applications — incorporating automated unit/integration testing, code quality gates, parallel build stages, artifact caching, automated rollback triggers, and staged rollouts — reducing deployment time by 50%, average build time by 30%, and release failure rate by 35%.
- Designed and deployed GitOps workflows using Argo CD, enabling declarative, version-controlled Kubernetes deployments with automatic drift detection and self-healing across Dev, QA, UAT, and Production environments.
- Spearheaded end-to-end MLOps platform development on Azure Machine Learning — architecting automated model training pipelines, model deployment workflows, model versioning via the AML Model Registry, experiment tracking with MLflow, and one-click model promotion workflows, including full model lifecycle management (registration, staging-to-production promotion, approval gates, and rollback) with complete audit traceability — reducing the data science team's release cycle from weeks to days.
- Built and operationalized a Py Spark-based distributed feature engineering pipeline processing 500GB+ daily datasets, enabling consistent feature computation for multiple ML models across training and inference environments.
- Implemented production model monitoring and data drift detection using Azure Monitor and custom Prometheus metrics — setting statistical thresholds on feature distributions and prediction confidence scores, triggering automated retraining pipelines on drift breach, reducing model degradation incidents by 60%.
- Designed and deployed REST API-based model deployment services for ML inference using Fast API and Gunicorn behind a NGINX reverse proxy — configuring ingress routing, TLS/SSL termination, and secure header defaults — containerized via Docker with optimized multi-stage Docker files (reducing image sizes by 40%), and served on AKS with Horizontal Pod Auto scaler, sustaining sub-100ms p99 latency at 1,000+ RPS with zero downtime deployments.
- Built AWS CDK stacks in Python to provision ECS Fargate clusters, VPC networking, IAM roles, security groups, and ALB configurations — replacing 2,000+ lines of CloudFormation with reusable, testable CDK constructs — and complemented this with Terraform and Ansible automation for EC2, S3, VPC, and IAM provisioning, cutting manual environment setup effort by 60%.
- Configured AWS ALB target groups, listener rules, and Route 53 health checks for ECS-backed services — enabling path-based routing and zero-downtime blue/green deployments — and published containerized Python and Node.js microservices to Azure Container Registry (ACR) as part of standardized release workflows.
- Integrated Hashi Corp Vault alongside Azure Key Vault and AWS Secrets Manager for dynamic secret injection, automated rotation, and least-privilege access across CI/CD pipelines and runtime workloads — eliminating hardcoded credentials.
- Embedded DevSecOps across all pipelines: integrated SonarQube for static code analysis, OWASP ZAP for dynamic security testing, Trivy for container image vulnerability scanning, and Checkov for IaC security policy enforcement — reducing critical CVEs by 40%.
- Managed and hardened Linux/Unix-based hosts (Ubuntu, RHEL) underpinning AWS EC2 and Azure VM workloads — applying OS-level security patches, SSH hardening, and CIS benchmark configurations via Bash scripting.
- Managed Kubernetes (AKS) clusters supporting 50+ microservices: implemented RBAC, Network Policies, Pod Disruption Budgets, Horizontal Pod Auto scaler, and Vertical Pod Auto scaler — maintaining 99.9% SLA across production workloads.
- Implemented centralized observability using a Prometheus + Grafana + Loki stack with custom dashboards and SLO/SLI definitions for ML model monitoring, infrastructure health, and pipeline SLAs, and configured Azure

Monitor alerts and Log Analytics workspaces for centralized log aggregation across 20+ microservices — reducing mean time to detect (MTTD) incidents by 45%.

- Led multi-environment release management (Dev → QA → UAT → Prod) using environment-specific Terraform workspaces, Helm value overrides, and pipeline promotion gates within Agile sprint cycles — ensuring zero-surprise production releases.
- Participated in on call/support rotation, driving root cause analysis (RCA) and permanent remediation for production incidents across Linux/Unix, Kubernetes, CI/CD, and cloud layers, while partnering closely with data science teams to debug ML training pipeline failures and automate dataset versioning via S3 lifecycle policies.
- Established infrastructure cost governance using AWS Cost Explorer and Azure Cost Management — identifying rightsizing opportunities and Reserved Instance purchases that reduced monthly cloud spend by 25%.
- Implemented Git branching strategies (Git Flow), enforced PR review policies, and integrated automated code linting and unit test gates into the merge pipeline — improving overall code quality across teams.
- Developed Python automation utilities and lightweight Flask/Fast API services for platform validation, smoke testing, and integration checks, with packaging and unit test coverage.
- Led the migration of 3 legacy monolithic applications to a Docker/Kubernetes microservices architecture on AWS ECS with Fargate — eliminating server management overhead, improving deployment frequency 4x, and reducing infrastructure cost by 20%.
- Maintained technical documentation, standard operating procedures (SOPs), and infrastructure runbooks in Confluence, enabling seamless knowledge transfer within the team.

KEY ACHIEVEMENTS

50% Faster Deployments	Engineered CI/CD pipelines across 15+ applications using Azure DevOps, GitHub Actions, and Jenkins (with shared libraries) — cutting deployment cycle time by 50%, average build time by 30%, and reducing release failures by 35% through automated testing, quality gates, and staged rollouts.
MLOps Platform Delivery	Built a production-grade MLOps platform on Azure ML with automated training pipelines, MLflow experiment tracking, model registry, drift detection, and REST inference APIs — reducing data science release cycles from weeks to days.
99.9% Production Uptime	Maintained 99.9% SLA across AKS production Kubernetes workloads through proactive monitoring (Prometheus/Grafana), SLO/SLI-driven alerting, HPA auto-scaling, Pod Disruption Budgets, and rapid on-call incident response with structured RCA.
40% CVE Reduction	Embedded a full DevSecOps toolchain (SonarQube, OWASP ZAP, Trivy, Checkov, Hashi Corp Vault) across all CI/CD pipelines — reducing critical application vulnerabilities by 40% and achieving compliance with SOC2 security requirements.
60% Fewer ML Incidents	Implemented data drift detection and model performance monitoring with automated retraining triggers — reducing ML model degradation incidents in production by 60%.
25% Cloud Cost Reduction	Identified rightsizing opportunities, Reserved Instance recommendations, and orphaned resource cleanup via AWS Cost Explorer & Azure Cost Management — reducing monthly cloud expenditure by 25%.
IaC at Scale	Authored 50+ reusable Terraform modules and AWS CDK constructs covering compute, networking, IAM, monitoring, and security — cutting manual infrastructure setup effort by 60% and eliminating configuration drift.
Microservices Migration	Led migration of 3 legacy monolithic applications to Docker/Kubernetes microservices on AWS ECS/Fargate — improving deployment frequency 4x, enhancing fault isolation, and reducing infra costs by 20%.
Observability Uplift	Deployed a centralized Prometheus + Grafana + Loki + Alert manager observability stack with custom ML pipeline dashboards and SLO/SLI tracking — reducing mean time to detect (MTTD) production incidents by 45%.
Team Enablement	Mentored 4 junior engineers through code reviews, pair programming, and workshops within Agile ceremonies — growing team velocity and establishing internal best practices for Terraform, Kubernetes, and MLOps design patterns.

KEY PROJECTS

Internal MLOps Platform

Azure Machine Learning • AKS • AWS • GCP • MLflow • Py Spark

- Designed and built a production-ready MLOps pipeline from the ground up on Azure Machine Learning, AWS, GCP, and Kubernetes — supporting end-to-end ML lifecycle from data ingestion to model deployment for 5+ active ML models in production.
- Architected automated ML training pipelines with parameterized compute clusters, data versioning, and environment isolation — enabling data scientists to trigger training runs via CI/CD without infrastructure involvement.
- Integrated MLflow for experiment tracking, parameter logging, and artifact storage — providing full reproducibility and enabling data science teams to compare 100+ experiments across model versions.
- Implemented automated model promotion workflows: models passing evaluation thresholds in staging were automatically promoted to production through pipeline approval gates, with rollback capability to previous versions.
- Built Py Spark-based distributed feature engineering pipelines processing 500GB+ daily data from multiple sources — computing and validating features consistently across training and inference to eliminate training-serving skew.
- Developed Fast API and Flask-based ML inference APIs with async request handling, containerized via Docker, and deployed on AKS with HPA auto-scaling — sustaining sub-100ms latency at 1,000+ RPS.
- Implemented model monitoring pipelines using Prometheus custom metrics and Grafana dashboards tracking prediction confidence, feature drift, and throughput, with SLO/SLI definitions and Alert manager notifications on drift breach.
- Integrated Amazon S3, Azure Blob Storage, and Google Cloud Storage for model artifact versioning and decoupled storage — separating artifact delivery from inference runtime to enable zero-downtime model hot-swapping.
- Built auto-scaling inference services using AWS ALB and Auto Scaling Groups (ASG) for burst traffic handling — achieving consistent low-latency serving even at 3x peak load.
- Containerized all ML training and inference workloads using Docker with reproducible environments, deployed on Kubernetes (AKS/GKE) with RBAC and Network Policies for security isolation.

DevOps & MLOps Engineer | Cyient

Delivered CI/CD automation, container orchestration, and MLOps workflows across AWS and Azure environments supporting data science and engineering teams within Agile sprints.

- Managed and hardened Linux/Unix-based servers (Ubuntu, RHEL) across AWS EC2 and Azure VMs for development and staging environments — applying OS-level security patches, SSH hardening, and CIS benchmark configurations via Bash scripting.
- Built and maintained Jenkins (including shared libraries) and Azure DevOps CI/CD pipelines for 5+ applications, introducing parallel build stages, artifact caching, and automated rollback triggers — improving average build time by 30%.
- Containerized ML workloads using Docker with optimized multi-stage Docker files — reducing image sizes by 40% and enabling consistent, reproducible model deployment across development, staging, and production environments.
- Designed model lifecycle management workflows covering model registration, environment promotion (staging → production), approval gates, and rollback procedures — providing full audit traceability for ML model releases.
- Authored Terraform and Ansible automation for provisioning AWS EC2 instances, S3 buckets, VPCs, security groups, and IAM roles — standardizing environment creation and reducing manual provisioning effort by 60%.
- Configured NGINX reverse proxy and ingress rules, including TLS certificate management, for internal services across staging and production.
- Implemented Git branching strategies (Git Flow) across multiple repositories, enforced PR review policies, and integrated automated code linting and unit test gates into the merge pipeline within Agile ceremonies — improving overall code quality.
- Collaborated closely with data science teams to debug ML training pipeline failures, optimize Docker-based training job environments, and automate dataset versioning using S3 lifecycle policies.
- Developed Python automation utilities and lightweight Flask/Fast API services for platform validation, smoke testing, and integration checks, with packaging and unit test coverage.

- Assisted in configuring Azure Monitor alerts and Log Analytics workspaces for centralized log aggregation across distributed microservices, improving cross-team debugging efficiency; performed RCA for recurring incidents.
- Supported AKS (Azure Kubernetes Service) cluster maintenance: applied Helm chart updates, managed namespace-level resource quotas, and performed rolling updates for test environment workloads.
- Containerized Node.js and Python microservices using Docker, authored multi-stage Docker files for optimized image builds, and published images to Azure Container Registry (ACR).
- Migrated legacy monolithic applications to microservices architecture deployed on AWS ECS with Fargate — eliminating server management overhead, improving horizontal scalability, and reducing infrastructure cost by 20%.
- Configured AWS ALB target groups, listener rules, and Route 53 health checks for ECS-backed services — enabling path-based routing and zero-downtime blue/green deployments.
- Implemented AWS CDK (Python) stacks to provision ECS clusters, VPC subnets, NAT gateways, IAM execution roles, security groups, and ALB configurations — replacing manual console-based provisioning.
- Maintained technical documentation, standard operating procedures (SOPs), and infrastructure runbooks in Confluence, enabling seamless knowledge transfer within the team.

DevOps / Cloud Engineer | Novelis

Azure & AWS Infrastructure Automation

- Managed hybrid Azure and AWS infrastructure for a global manufacturing enterprise — automating environment provisioning, deployment pipelines, and operational monitoring across 3 regions.
- Authored Terraform modules for provisioning Azure VMs, Virtual Networks, NSGs, Storage Accounts, and Azure Key Vault — replacing ad-hoc portal-based setup and reducing environment creation time by 50%.
- Designed and implemented CI/CD build and deployment pipelines using Azure Pipelines with environment-specific variable groups, approval gates, and automated rollback — accelerating release cycles by 40%.
- Configured Linux and RHEL VMs in Azure with hardened OS configurations, automated patching via Azure Update Management, and CIS benchmark compliance — ensuring zero security audit findings.
- Deployed Prometheus and Grafana monitoring with custom alerting for infrastructure KPIs (CPU, memory, disk, network) and application health endpoints — reducing incident detection time from hours to minutes.
- Secured all sensitive credentials and connection strings using Azure Key Vault with managed identity access — eliminating all hardcoded secrets and achieving zero-secret exposure policy compliance.
- Implemented Azure Monitor alerts and Log Analytics dashboards for centralized logging — aggregating logs from 20+ VMs and services for cross-team operational visibility.

DevOps Engineer | Micron

Kubernetes & CI/CD Automation

- Deployed and managed containerized microservices using Docker and Kubernetes (AKS) — achieving 99.9% uptime across 15+ production services through rolling updates, readiness probes, and pod disruption budgets.
- Automated full CI/CD workflows using Azure DevOps, Azure Pipelines, and GitHub Actions — integrating build, test, SAST scanning, container image build/push to ACR, and Kubernetes rolling deployment stages — reducing release cycle time by 50%.
- Integrated SonarQube for static code analysis with quality gate enforcement blocking merges on critical issues, and Trivy for container image scanning — reducing security vulnerabilities by 30%.
- Managed Helm chart releases across multiple environments using Helm files and environment-specific value overrides — standardizing Kubernetes application deployment and eliminating manual manifest management.
- Implemented Azure Key Vault integration with AKS using pod-managed identities and the Secrets Store CSI driver — enabling secure dynamic secret injection without code changes.
- Configured Horizontal Pod Auto scaler (HPA) and Cluster Auto scaler on AKS — enabling automatic scale-out during traffic spikes and scale-in during off-peak hours, reducing compute costs by 20%.
- Built and maintained CI/CD pipelines for 10+ microservices with independent deployment cadences — enabling teams to release independently without cross-team coordination bottlenecks.

EDUCATION & CERTIFICATIONS

B. Com in Computer Science — Acharya Nagarjuna University | 2016 – 2019